SMOOTHNESS IS EQUIVALENT TO LIFTING ALONG SMALL EXTENSIONS OF ARTINIAN LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3623-3641 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3643-3665, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smooth morphisms, small extensions of Artinian local rings, and the infinitesimal lifting criterion; the converse is retained.

Source reference. Original lemma lemma-lifting-along-artinian-at-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3575–3600. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-smooth-formally-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3190–3197. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-artinian-finite-type: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3665–3665. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-section-points-finite-type: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3665–3665. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume that S is locally Noetherian and f locally of finite type. The following are equivalent:*

(1) *f is smooth,*

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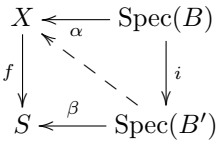
(2) *for every solid commutative diagram*



where $B' \to B$ is a small extension of Artinian local rings and β of finite type (!) there exists a dotted arrow making the diagram commute.

Proof. If f is smooth, then the infinitesimal lifting criterion (Lemma lemma-smooth-formally-smooth) says f is formally smooth and (2) holds.

Assume (2). The set of points $x \in X$ where f is not smooth forms a closed subset T of X. By the discussion in Morphisms, Section morphisms-section-points-finite-type, if $T \neq \emptyset$ there exists a point $x \in T \subset X$ such that the morphism

$$\mathrm{Spec}(\kappa(x)) \to X \to S$$

is of finite type (namely, pick any point x of T which is closed in an affine open of X). By Morphisms, Lemma morphisms-lemma-artinian-finite-type given any local Artinian ring B' with residue field $\kappa(x)$ then any morphism $\beta : \mathrm{Spec}(B') \to S$ is of finite type. Thus we see that all the diagrams used in Lemma lemma-lifting-along-artinian-at-point (4) correspond to diagrams as in the current lemma (2). Whence $X \to S$ is smooth a x a contradiction. □